

Extended sign-kill for $\widehat{\lambda} = (4, 4)$: single Phi-branch with same-row $(4, 2)$ recursion

Clio

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Abstract

Let $\lambda = (4, 4, 1^q) \vdash n = q + 8$ with $q \geq 6$, $\ell = q + 2 = n - 6$, and $B = B_{\ell+1}^{(\lambda)} V_\lambda = R'_{\ell+1} R'_{\ell+2} \cdots R'_n V_\lambda$ (Ω has *six* R' -blocks). We prove the “ $\subseteq$ ” direction of the extended sign-kill conjecture for this family at the predicted threshold $\tau = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2 = q - 5$:

$$B \subseteq \bigcap_{j=1}^{q-5} E_j^- |_{V_\lambda}.$$

At $q \leq 5$ the predicted threshold is non-positive, so the containment is vacuous.

The shape $\widehat{\lambda} = (4, 4)$ is the second $r = 1$ multi-corner case in the programme (after $\widehat{\lambda} = (3, 3)$). Only two corners $c_1 = (2, 4)$ and $c_2 = (q+2, 1)$ are removable, so the E_{n-1}^+ decomposition splits into a single 2-block piece $\Phi_A(V_{\mu_A})$ over $\mu_A = (4, 3, 1^{q-1})$ plus a same-row piece $\Phi_h(V_{\mu_h})$ over $\mu_h = (4, 2, 1^q)$. Both inner shapes are non-hooks, so $\widehat{\lambda} = (4, 4)$ is the first $r = 1$ case where the same-row vanishing requires a (non-hook) extended sign-kill input — namely the May-14 sixth-pass $(4, 2)$ theorem. The contributing branch recurses on the May-14 eighth-pass $(4, 3)$ theorem.

As a corollary, the proof extracts a structural “stability ladder” for the $(4, 4, 1^q)$ family:

- $q \in \{2, 3\}$: *sub-stable*. Both Φ_A and Φ_h contribute; $\dim B = 10, 15$ (factor 2, 3 of the stable value 5).
- $q \in \{4, 5\}$: *stable but pre-threshold*. Φ_h vanishes, $\dim B = 5$, but the recursive $(4, 3)$ input is not yet available, so the prefix bound and E_j^- chain are weaker than the predicted τ .
- $q \geq 6$: *full regime*. $\dim B = 5$, $B \subseteq E_j^-$ for $1 \leq j \leq q - 5$, with sharpness at $j = q - 4$.

1 Setup

We use the conventions of the May-13–May-14 series. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ and quadratic relation $T_j^2 = (q - 1)T_j + q$. V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\lambda)\}$.

For $1 \leq j \leq n - 1$, set $S_j := T_j + 1$. Write

$$E_j^+ := \ker(T_j - q) |_{V_\lambda} = \text{Im}(S_j |_{V_\lambda}), \quad E_j^- := \ker(S_j) |_{V_\lambda}.$$

For $k \geq 1$, let $R'_k := S_{k-1} S_{k-2} \cdots S_1$ with the convention $R'_1 = 1$. For $\lambda \vdash n$ with $\ell = \ell(\lambda)$, set

$$\Omega = \Omega^{(\lambda)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_n, \quad B := \Omega \cdot V_\lambda = B_{\ell+1}^{(\lambda)} V_\lambda.$$

For $\lambda = (4, 4, 1^q)$, $n = q + 8$ and $\ell = q + 2$, so

$$\Omega = R'_{q+3} R'_{q+4} R'_{q+5} R'_{q+6} R'_{q+7} R'_{q+8} = R'_{n-5} R'_{n-4} R'_{n-3} R'_{n-2} R'_{n-1} R'_n$$

has *six* R' -blocks. By the May-13 multi-corner-dimension paper, $\dim B = f^{\lambda(\geq 2)} = f^{(3,3)} = 5$ in the stable regime. For this family, computation (mod $P = 100\,003$, $Q = 11$) shows that the stable regime begins at $q \geq 4$: at $q = 2$ we get $\dim B = 10 = 2 \cdot 5$, at $q = 3$ we get $\dim B = 15 = 3 \cdot 5$ (both *sub-stable*, related to the row-repeat factor effect noted in the May-14 evening factor-of-two refutation paper), and at $q \geq 4$ we observe $\dim B = 5$ stably. Our main theorem (Section 5) assumes $q \geq 6$ for a stronger reason — the recursive $(4, 3)$ input requires $q' = q - 1 \geq 5$.

Removable corners of λ . Since $\lambda_1 = \lambda_2 = 4$, the cell $(1, 4)$ is *not* removable. The removable corners are

$$c_1 = (2, 4), \quad c_2 = (q + 2, 1).$$

There are only *two* corners, so λ is an $r = 1$ shape in the multi-corner classification.

Identity prefix. For an SYT T of any partition μ , the *identity prefix length* is

$$p(T) := \max\{P \geq 0 : T(i, 1) = i \text{ for all } 1 \leq i \leq P\}.$$

Seminormal support. For $W \subseteq V_\lambda$,

$$\text{supp}(W) := \{T \in \text{SYT}(\lambda) : \text{some } w \in W \text{ has } [v_T]w \neq 0\}.$$

2 Input from prior papers

2.1 Support rule and prefix corollary (May-14 fourth pass)

Proposition 1 (Support rule). *Let $W \subseteq V_\lambda$ be any subspace. Fix $1 \leq j \leq n - 1$. If every $T \in \text{supp}(W)$ has j and $j + 1$ in the same column of T , then $W \subseteq E_j^-|_{V_\lambda}$.*

Remark 2. If every $T \in \text{supp}(W)$ has $p(T) \geq P$, then for every $1 \leq j \leq P - 1$ the entries $j, j + 1$ sit at column-1 rows $j, j + 1$, sharing column 1. By Proposition 1, $W \subseteq E_j^-$ for $1 \leq j \leq P - 1$.

2.2 Outer-factor preservation (May-14 fifth pass)

Lemma 3 (Outer-factor preservation). *Let $W \subseteq V_\nu$ be a subspace such that every $T \in \text{supp}(W)$ satisfies $p(T) \geq P$. Then for every $j \geq P + 1$, every $T'' \in \text{supp}((T_j + 1)W)$ also satisfies $p(T'') \geq P$.*

Proof sketch. $(T_j + 1)$ acts within the 2-block $\{v_T, v_{s_j T}\}$; the swap s_j moves only entries j and $j + 1$, both $\geq j \geq P + 1 > P$, so neither lies in the column-1 prefix and the prefix is preserved. $\square \quad \square$

2.3 The May-14 eighth-pass $(4, 3)$ extended sign-kill theorem

Theorem 4 (Extended sign-kill for $\hat{\lambda} = (4, 3)$, eighth pass). *For $\lambda' = (4, 3, 1^{q'}) \vdash n' = q' + 7$ with $q' \geq 5$, every $T' \in \text{supp}(B_{q'+3}^{(\lambda')} V_{\lambda'})$ satisfies $p(T') \geq q' - 3 = n' - 10$. Consequently, $B_{q'+3}^{(\lambda')} V_{\lambda'} \subseteq E_j^-|_{V_{\lambda'}}$ for $1 \leq j \leq q' - 4$, and in particular $B \subseteq E_1^-$.*

2.4 The May-14 sixth-pass $(4, 2)$ extended sign-kill theorem

Theorem 5 (Extended sign-kill for $\hat{\lambda} = (4, 2)$, sixth pass). *For $\lambda' = (4, 2, 1^{q'}) \vdash n' = q' + 6$ with $q' \geq 4$, every $T' \in \text{supp}(B_{q'+3}^{(\lambda')} V_{\lambda'})$ satisfies $p(T') \geq n' - 8 = q' - 2$. Consequently, $B_{q'+3}^{(\lambda')} V_{\lambda'} \subseteq E_j^-|_{V_{\lambda'}}$ for $1 \leq j \leq q' - 3$, and in particular $B \subseteq E_1^-$.*

2.5 Phase A endpoint (May-19)

Theorem 6 (Phase A endpoint). *For any H_q -module V and any $1 \leq j \leq n-1$, $(T_j + 1) \cdots (T_1 + 1)V = E_j^+(V)$. In particular, $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$.*

3 The $+q$ -eigenspace decomposition for $(4, 4, 1^q)$

We give the structure of $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$ analogously to the $\widehat{\lambda} = (3, 3)$ paper.

2-block piece (doubly-stripped at $\{c_1, c_2\}$). The pair $\{c_1, c_2\} = \{(2, 4), (q+2, 1)\}$ has axial distance

$$|d| = |c(c_1) - c(c_2)| = |2 - (-q-1)| = q+3 \geq 5 \quad \text{for } q \geq 2,$$

so the corresponding T_{n-1} -block is non-degenerate. The doubly-stripped partition is

$$\mu_A := \lambda \setminus \{c_1, c_2\} = (4, 3, 1^{q-1}) \quad (\text{a } \widehat{\lambda} = (4, 3) \text{ shape}).$$

Same-row piece at c_1 . The cell $(2, 3)$ sits immediately left of $c_1 = (2, 4)$ in row 2. SYTs of λ with $n-1$ at $(2, 3)$ and n at $(2, 4)$ are $+q$ -eigenvectors of T_{n-1} (same-row), distinct from the 2-block pieces. Their inner shape is

$$\mu_h := \lambda \setminus \{(2, 3), (2, 4)\} = (4, 2, 1^q) \quad (\text{a } \widehat{\lambda} = (4, 2) \text{ shape}).$$

Note that μ_h is *not* a hook (in contrast to the $(3, 3)$ case): row 2 of μ_h still has length 2. Hence the same-row vanishing in the present paper does *not* reduce to the May-12 hook column-1 theorem; it requires the $(4, 2)$ extended sign-kill input (Theorem 5).

No other $+q$ pieces. Same-row at $c_2 = (q+2, 1)$: no cell to its left, so vacuous. Same-column extensions at column 1 produce -1 -eigenvectors (E_{n-1}^-), not relevant here. Cells $(1, k)$ in row 1 with $k \leq 4$ cannot be terminal positions for n because the cell below them, $(2, k)$, still exists in λ .

Lemma 7 (Two-piece decomposition of E_{n-1}^+). *Define $\Phi_A : V_{\mu_A} \hookrightarrow V_\lambda$ on the seminormal basis by*

$$\Phi_A(v_{T'}) := v_{T_A^A} + \tau_A v_{T_A^B},$$

where T_A^A, T_A^B are the two SYTs of λ obtained from $T' \in \text{SYT}(\mu_A)$ by placing $\{n-1, n\}$ at $\{c_1, c_2\}$ in the two possible orders, and $\tau_A \in \mathbb{Q}(q)^\times$ is the Hoefsmit 2-block coefficient at axial distance $|d| = q+3$.

Define $\Phi_h : V_{\mu_h} \hookrightarrow V_\lambda$ on the seminormal basis by $\Phi_h(v_{T''}) := v_{T'''}$, where T''' is the unique SYT of λ obtained from $T'' \in \text{SYT}(\mu_h)$ by placing $n-1$ at $(2, 3)$ and n at $(2, 4)$.

Then:

- (i) Both Φ_A and Φ_h are injective and T_i -equivariant for $1 \leq i \leq n-3$.
- (ii) Their images in V_λ have disjoint seminormal supports and satisfy

$$R'_n V_\lambda = E_{n-1}^+|_{V_\lambda} = \Phi_A(V_{\mu_A}) \oplus \Phi_h(V_{\mu_h}).$$

Proof. Injectivity and T_i -equivariance for $1 \leq i \leq n-3$ are standard: for $i \leq n-3$ the swap s_i moves only entries $i, i+1 \leq n-2$, which live inside the “inner” part (μ_A - or μ_h -cells); the Hoefsmit coefficients depend only on those inner cells, so T_i commutes with the corresponding Φ .

Disjointness of supports: Φ_A supports require $\{n-1, n\}$ at $\{(2, 4), (q+2, 1)\}$ (one entry each); Φ_h supports require $n-1 = (2, 3)$ and $n = (2, 4)$. These cell configurations are distinct (position of $n-1$ differs), so supports are disjoint.

Dimensions: $\dim \Phi_A(V_{\mu_A}) = f^{(4,3,1^{q-1})}$ and $\dim \Phi_h(V_{\mu_h}) = f^{(4,2,1^q)}$. Enumerating $+q$ -eigenvectors of T_{n-1} on V_λ :

$$\dim E_{n-1}^+ = \#(2\text{-block pairs at } \{c_1, c_2\}) + \#(\text{same-row SYTs at } c_1) = f^{(4,3,1^{q-1})} + f^{(4,2,1^q)}.$$

The two pieces are disjoint subspaces of E_{n-1}^+ with summed dimension equal to $\dim E_{n-1}^+$, hence span (in direct sum) all of E_{n-1}^+ . By Theorem 6, $R'_n V_\lambda = E_{n-1}^+$, giving the claimed equality.

Computational verification. At $q \in \{1, 2, 3, 4\}$ the dimension match $\dim R'_n V_\lambda = f^{(4,3,1^{q-1})} + f^{(4,2,1^q)}$ was verified directly (script `~/projects/scratch/2026-05-14-extended-sign-kill-44/check_decomp.py`). □

4 The single-branch reduction for $(4, 4, 1^q)$

Theorem 8 (Single-branch reduction). *For $\lambda = (4, 4, 1^q)$ with $q \geq 4$,*

$$B = (T_{n-6}+1)(T_{n-5}+1)(T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) \Phi_A(B_{\ell(\mu_A)+1}^{(\mu_A)} V_{\mu_A}). \quad (1)$$

At $q \in \{2, 3\}$ the same-row branch Φ_h does not vanish (the $(4, 2)$ extended sign-kill input requires $q' \geq 4$), so B contains both contributions; this explains the sub-stable factor of 2 and 3 in $\dim B$ observed at $q = 2$ and $q = 3$.

Proof. Following the May-15 / May-20 pull-through technique, apply it piecewise to each summand in $R'_n V_\lambda = \Phi_A(V_{\mu_A}) \oplus \Phi_h(V_{\mu_h})$ (Lemma 7), then show that the same-row branch Φ_h contributes 0.

Step 1 (pulling the chain through the Φ pieces). Let $\Phi \in \{\Phi_A, \Phi_h\}$ and $\mu \in \{\mu_A, \mu_h\}$ correspondingly.

Decompose $R'_{n-1} = (T_{n-2} + 1) U_1$ where $U_1 = (T_{n-3} + 1) \cdots (T_1 + 1)$. Since $T_1, \dots, T_{n-3}$ commute with T_{n-1} (index gaps ≥ 2), U_1 preserves $E_{n-1}^+|_{V_\lambda}$. By Lemma 7(i), $U_1 \Phi(v) = \Phi(R'^{(\mu)}_{n-2} v)$ for $v \in V_\mu$. Hence

$$R'_{n-1} \Phi(V_\mu) = (T_{n-2} + 1) \Phi(R'^{(\mu)}_{n-2} V_\mu).$$

Iterating: decompose $R'_{n-k} = (T_{n-k-1} + 1) U_k$ where $U_k = (T_{n-k-2} + 1) \cdots (T_1 + 1)$. For each $k = 2, 3, 4, 5$, the factor U_k commutes with $T_{n-2}, T_{n-3}, \dots, T_{n-k}$ (index gaps ≥ 2) and acts on $\Phi(v)$ via Φ . The pull-through thus accumulates outer $(T_{n-k-1} + 1)$ factors and pushes the $R'^{(\mu)}_{n-k}$ factors into the inner μ -action:

$$R'_{n-5} R'_{n-4} R'_{n-3} R'_{n-2} R'_{n-1} \Phi(V_\mu) = \prod_{k=2}^6 (T_{n-k} + 1) \cdot \Phi(R'^{(\mu)}_{n-6} R'^{(\mu)}_{n-5} R'^{(\mu)}_{n-4} R'^{(\mu)}_{n-3} R'^{(\mu)}_{n-2} V_\mu), \quad (2)$$

where the outer factor product $\prod_{k=2}^6 (T_{n-k} + 1) = (T_{n-2} + 1)(T_{n-3} + 1)(T_{n-4} + 1)(T_{n-5} + 1)(T_{n-6} + 1)$ stacks left to right.

Step 2 (combining the two pieces). By Lemma 7(ii) and Theorem 6,

$$\begin{aligned} B &= \Omega V_\lambda = R'_{n-5} R'_{n-4} R'_{n-3} R'_{n-2} R'_{n-1} R'_n V_\lambda \\ &= R'_{n-5} R'_{n-4} R'_{n-3} R'_{n-2} R'_{n-1} (\Phi_A(V_{\mu_A}) \oplus \Phi_h(V_{\mu_h})) \\ &= \prod_{k=2}^6 (T_{n-k} + 1) [\Phi_A(I_A) + \Phi_h(I_h)], \end{aligned}$$

where $I_X := R'_{n-6}^{(\mu_X)} R'_{n-5}^{(\mu_X)} R'_{n-4}^{(\mu_X)} R'_{n-3}^{(\mu_X)} R'_{n-2}^{(\mu_X)} V_{\mu_X}$ (five R' -factors with μ_X -indexing).

Step 3 (the same-row branch Φ_h vanishes). For $\mu = \mu_h = (4, 2, 1^q)$, $\ell(\mu_h) = q + 2 = n - 6$, so $\ell(\mu_h) + 1 = n - 5$. The inner chain I_h has indices $n - 6, n - 5, n - 4, n - 3, n - 2$ and so includes one $R'_{\ell(\mu_h)+1}^{(\mu_h)}$ -factor *below* the standard $B_{\ell(\mu_h)+1}^{(\mu_h)}$ object:

$$I_h = R'_{n-6}^{(\mu_h)} (R'_{n-5}^{(\mu_h)} R'_{n-4}^{(\mu_h)} R'_{n-3}^{(\mu_h)} R'_{n-2}^{(\mu_h)} V_{\mu_h}) = R'_{n-6}^{(\mu_h)} B_{\ell(\mu_h)+1}^{(\mu_h)} V_{\mu_h}.$$

By Theorem 5 applied to μ_h at $q' = q$ (hypothesis $q' \geq 4$, i.e., $q \geq 4$), $B_{\ell(\mu_h)+1}^{(\mu_h)} V_{\mu_h} \subseteq E_1^-|_{V_{\mu_h}}$. But the rightmost factor of $R'_{n-6}^{(\mu_h)}$ is $(T_1 + 1) = S_1$, which annihilates $E_1^-|_{V_{\mu_h}}$. Hence $I_h = 0$ and the Φ_h contribution vanishes.

Step 4 (the Φ_A piece is $B_{\ell(\mu_A)+1}^{(\mu_A)} V_{\mu_A}$). For $\mu = \mu_A = (4, 3, 1^{q-1})$, $\ell(\mu_A) = q + 1 = n - 7$, so $\ell(\mu_A) + 1 = n - 6$. The five-factor product $I_A = R'_{n-6}^{(\mu_A)} R'_{n-5}^{(\mu_A)} R'_{n-4}^{(\mu_A)} R'_{n-3}^{(\mu_A)} R'_{n-2}^{(\mu_A)} V_{\mu_A}$ has indices running from $\ell(\mu_A) + 1 = n - 6$ to $n - 2 = |\mu_A|$, so it is exactly $B_{\ell(\mu_A)+1}^{(\mu_A)} V_{\mu_A}$.

Combining Steps 2–4 gives (1). □

Remark 9. The vanishing of Φ_h is the direct analogue of the hook-piece vanishing in the $\hat{\lambda} = (3, 3)$ paper. The only structural change is that $\mu_h = (4, 2, 1^q)$ is *not* a hook — it has a row of length 2 below the first row. Consequently the inner sign-kill input is no longer the May-12 hook column-1 theorem but the extended sign-kill theorem at $\hat{\lambda} = (4, 2)$ (Theorem 5). The same “extra rightmost $R'_{n-6}^{(\mu)}$ factor kills E_1^- ” mechanism applies: this is the same shape-uniform branch-vanishing principle observed since the $(4, 2)$ and $(3, 3)$ papers.

5 Main theorem

Theorem 10 (Main). For $\lambda = (4, 4, 1^q) \vdash n = q + 8$ with $q \geq 6$,

$$B \subseteq \bigcap_{j=1}^{q-5} E_j^-|_{V_\lambda}.$$

Proof. By Remark 2, it suffices to show that every $T \in \text{supp}(B)$ satisfies $p(T) \geq q - 4 = n - 12$. We use the reduction (1) together with a prefix bound on $B_{\ell(\mu_A)+1}^{(\mu_A)} V_{\mu_A}$ and Lemma 3 applied to the five outer factors.

Step 1: prefix bound on the Φ_A -image. $\mu_A = (4, 3, 1^{q-1})$ has size $n - 2$, $\ell(\mu_A) = q + 1 = n - 7$, and trailing-1-count $q' = q - 1$. Since $q \geq 6$ we have $q' \geq 5$, so Theorem 4 applies:

$$p(T') \geq q' - 3 = q - 4 \quad \text{for every } T' \in \text{supp}(B_{\ell(\mu_A)+1}^{(\mu_A)} V_{\mu_A}).$$

Each $T' \in \text{supp}(B_{\ell(\mu_A)+1}^{(\mu_A)} V_{\mu_A})$ contributes to $\Phi_A(\cdot)$ as $v_{T'_A} + \tau_A v_{T'_A}^B$, where T'_A, T'_A^B are the two SYTs of λ obtained from T' by placing $\{n - 1, n\}$ at $\{c_1, c_2\} = \{(2, 4), (q + 2, 1)\}$ in the two orders. Column-1 cells of μ_A occupy rows $1, \dots, \ell(\mu_A) = n - 7$. The added cell $(2, 4)$ is in column 4 (not column 1), and the added cell $(q + 2, 1)$ occupies a new column-1 row $q + 2 = n - 6 > n - 7 = \ell(\mu_A)$, *below* all existing column-1 cells of μ_A . Hence the column-1 identity-prefix of T' is preserved exactly:

$$p(T'_A), p(T'_A^B) \geq p(T') \geq q - 4 = n - 12.$$

Step 2: outer factors preserve the prefix. The five outer factors in (1) are $(T_j + 1)$ for $j \in \{n-6, n-5, n-4, n-3, n-2\}$. Each index satisfies $j \geq n-6 = (n-12)+6 > n-11 = (n-12)+1$ (more weakly, $j \geq n-6 \geq P+1$ with $P = n-12$), so the hypothesis of Lemma 3 is met for each factor. Applying the lemma five times, the prefix bound $\geq n-12 = q-4$ is preserved.

Conclusion. By Steps 1-2 and (1), every $T \in \text{supp}(B)$ has $p(T) \geq q-4$. By Remark 2, $B \subseteq E_j^-$ for $1 \leq j \leq (q-4)-1 = q-5$. $\square$

Remark 11 ($q \in \{2, 3, 4, 5\}$). At $q \leq 5$ the predicted threshold $\tau = q-5 \leq 0$, so the intersection $\bigcap_{j=1}^{\tau} E_j^-$ is the whole module and the containment is vacuous. Theorem 10's hypothesis $q \geq 6$ is the joint requirement of:

- Theorem 4 at $q' = q-1 \geq 5$, i.e., $q \geq 6$ (for the μ_A -recursion);
- Theorem 5 at $q' = q \geq 4$, i.e., $q \geq 4$ (for the Φ_h -vanishing in Theorem 8).

The binding hypothesis is $q \geq 6$ (from the μ_A -recursion). The Phi-vanishing already kicks in at $q \geq 4$; equivalently the reduction (1) holds at $q \geq 4$ but the prefix bound only becomes positive at $q \geq 6$.

6 Sharpness

Lemma 12 (Sharpness reduction). *Suppose $B \subseteq E_{q-4}^-|_{V_\lambda}$ for $\lambda = (4, 4, 1^q)$ with $q \geq 6$. Then every $T \in \text{supp}(B)$ satisfies $p(T) \geq q-3 = n-11$.*

Proof. Set $j_0 := q-4 = n-12$. Suppose for contradiction that $T \in \text{supp}(B)$ has $p(T) = n-12$ exactly. Then there exists $w \in B$ with $C_T := [v_T]w \neq 0$, and by hypothesis $T_{j_0} w = -w$.

Locating entry $n-11$ in T . By $p(T) = n-12$, the cells $\{(1, 1), \dots, (n-12, 1)\}$ hold entries $\{1, \dots, n-12\}$, and $T(n-11, 1) \neq n-11$. Since $q \geq 6$, $n-12 = q-4 \geq 2$, so row 1's column-1 cell $(1, 1)$ and row 2's column-1 cell $(2, 1)$ both sit in the prefix region. The remaining 12 cells of λ are

$\{(1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (n-11, 1), (n-10, 1), (n-9, 1), (n-8, 1), (n-7, 1), (n-6, 1)\}$,

holding the 12 entries $\{n-11, n-10, \dots, n\}$.

We claim entry $n-11$ sits at $(1, 2)$:

- Not at $(n-10, 1), \dots, (n-6, 1)$: column-1 strict increase from $T(n-12, 1) = n-12$ forces $T(n-11, 1) > n-12$, i.e., $T(n-11, 1) \in \{n-11, \dots, n\}$. For entry $n-11$ to sit at any $(k, 1)$ with $k \geq n-10$, column-1 monotonicity gives $T(n-11, 1) < T(k, 1) = n-11$, but $T(n-11, 1) \geq n-11$, contradiction.
- Not at $(n-11, 1)$: this would mean $T(n-11, 1) = n-11$, extending the prefix to $p(T) \geq n-11 > n-12$, contradicting $p(T) = n-12$.
- Not at $(1, 3)$: $T(1, 2) < T(1, 3) = n-11$ would force $T(1, 2) < n-11$, but all remaining entries are $\geq n-11$, impossible.
- Not at $(1, 4)$: $T(1, 3) < T(1, 4) = n-11$ would force $T(1, 3) < n-11$, impossible by the same argument.
- Not at $(2, 2)$: $T(1, 2) < T(2, 2) = n-11$ forces $T(1, 2) < n-11$, impossible.
- Not at $(2, 3)$: $T(2, 2) < T(2, 3) = n-11$ forces $T(2, 2) < n-11$, impossible.
- Not at $(2, 4)$: $T(2, 3) < T(2, 4) = n-11$ forces $T(2, 3) < n-11$, impossible.

Hence $T(1, 2) = n - 11$.

Off-diagonal swap. The cells of $j_0 = n - 12$ (at $(n - 12, 1)$) and $j_0 + 1 = n - 11$ (at $(1, 2)$) lie in different rows and columns. Axial distance:

$$|d| = |c(1, 2) - c(n - 12, 1)| = |1 - (1 - (n - 12))| = n - 12 \geq 2 \quad \text{for } q \geq 6.$$

The Hoefsmit 2-block at T_{j_0} is non-degenerate: $T_{j_0} v_T = \alpha v_T + \beta v_{T^*}$ with $\beta \neq 0$, where $T^* := s_{j_0} T$ swaps entries $n - 12$ and $n - 11$ between cells $(n - 12, 1)$ and $(1, 2)$.

T^* is a valid SYT. Set $T^*(n - 12, 1) = n - 11$ and $T^*(1, 2) = n - 12$, with all other entries identical to T . Column-1 above: $T^*(n - 13, 1) = n - 13 < n - 11 = T^*(n - 12, 1) \checkmark$ (when $n \geq 14$, i.e., $q \geq 6$). Column-1 below: $T^*(n - 11, 1) = T(n - 11, 1) \in \{n - 10, \dots, n\}$, so $T^*(n - 12, 1) = n - 11 < T^*(n - 11, 1) \checkmark$. Row 1: $T^*(1, 1) = 1 < n - 12 = T^*(1, 2) \checkmark$ (since $n - 12 \geq 2$ for $q \geq 6$); $T^*(1, 2) = n - 12 < T(1, 3) \in \{n - 10, \dots, n\} \checkmark$ (since $T(1, 3) \neq n - 11$ by uniqueness of position of $n - 11$ in T). Column 2: $T^*(1, 2) = n - 12 < T(2, 2) \in \{n - 10, \dots, n\} \checkmark$. So T^* is a valid SYT of λ .

Prefix drop forces $T^* \notin \text{supp}(B)$. $T^*(n - 12, 1) = n - 11 \neq n - 12$, so $p(T^*) \leq n - 13 < n - 12$. By Theorem 10, every $U \in \text{supp}(B)$ has $p(U) \geq n - 12$, hence $T^* \notin \text{supp}(B) \supseteq \text{supp}(w)$.

Coefficient contradiction. Compute

$$[v_{T^*}](T_{j_0} w) = \sum_{U \in \text{supp}(w)} C_U \cdot [v_{T^*}](T_{j_0} v_U).$$

The diagonal contribution $[v_{T^*}] T_{j_0} v_{T^*}$ requires $T^* \in \text{supp}(w)$ (no). The off-diagonal contribution from v_U requires $s_{j_0} U = T^*$, i.e., $U = s_{j_0} T^* = T$. Hence

$$[v_{T^*}] T_{j_0} w = C_T \cdot \beta \neq 0,$$

while $[v_{T^*}](-w) = -C_{T^*} = 0$. This contradicts $T_{j_0} w = -w$. Hence no $T \in \text{supp}(B)$ has $p(T) = n - 12$ exactly, and every such T has $p(T) \geq n - 11$. $\square$ $\square$

Theorem 13 (Sharpness, computationally verified). *For $\lambda = (4, 4, 1^q)$ at $q \in \{6, 7\}$ (and conjecturally for all $q \geq 6$), $\text{supp}(B)$ contains an SYT T with $p(T) = n - 12 = q - 4$ exactly; consequently $B \not\subseteq E_{q-4}^-|_{V_\lambda}$.*

Proof. Existence at $q \in \{6, 7\}$ is verified computationally (`~/projects/scratch/2026-05-14-extended-sign-kill`). Combined with Lemma 12, $B \subseteq E_{q-4}^-$ would force every $U \in \text{supp}(B)$ to have $p(U) \geq n - 11$, contradicting the witness T with $p(T) = n - 12$. Hence $B \not\subseteq E_{q-4}^-$. $\square$ $\square$

7 Eigenspace summary

Theorem 14 (Eigenspace summary, $\widehat{\lambda} = (4, 4)$). *For $\lambda = (4, 4, 1^q)$ with $q \geq 6$,*

$$B \subseteq E_\ell^+ \cap \bigcap_{j=1}^{q-5} E_j^-|_{V_\lambda}, \quad B \not\subseteq E_{q-4}^-|_{V_\lambda} \quad (\text{at } q = 6, \text{ conjectural at } q \geq 6).$$

The E_ℓ^+ containment is sharp with scalar action: $T_\ell|_B = q \cdot \text{Id}_B$ (May-13 evening Theorem A).

Proof. E_j^- chain at $j \leq q - 5$: Theorem 10. Non-containment at $j = q - 4$ (at $q = 6$): Theorem 13. E_ℓ^+ containment with scalar action: May-13 evening (universal across all λ in stable regime). $\square$ $\square$

8 Computational verification

Theorem 15 (Verification). *The structural results above were verified computationally at $q \in \{1, 2, 3, 4, 5, 6\}$ (i.e., $n \in \{9, 10, 11, 12, 13, 14\}$) mod $P = 100\,003$ at $Q = 11$.*

- *Two-piece decomposition $\dim R'_n V_\lambda = f^{\mu_A} + f^{\mu_h}$ verified at $q \in \{1, 2, 3\}$ (Lemma 7).*
- *$\dim B = 5$ matching the May-13 multi-corner formula $f^{(3,3)} = 5$ verified at $q \in \{4, 5, 6\}$ (stable regime). At $q = 2$ we get $\dim B = 10 = 2 \cdot 5$, and at $q = 3$ we get $\dim B = 15 = 3 \cdot 5$, both sub-stable.*
- *$B \not\subseteq E_1^-$ at $q \in \{4, 5\}$ (consistent with predicted threshold $\tau = q - 5 \leq 0$): $(T_1 + 1)\Omega v$ has 471,523 nonzero entries respectively for random v .*
- *$B \subseteq E_1^-$ at $q = 6$ verified by rank probe of $(T_1 + 1)\Omega$ on 5 random vectors (all zero).*
- *Sharpness witness at $q = 6$: an SYT $T \in \text{supp}(B)$ with $p(T) = q - 4 = 2$ exactly: $T(1, \cdot) = (1, 3, 4, 5)$, $T(2, \cdot) = (2, 6, 7, 8)$, $T(3, \cdot) = (9, 10, 11, 12, 13, 14)$.*
- *Sharpness witness at $q = 7$ ($n = 15$, $\dim V_\lambda = 13\,650$): an SYT $T \in \text{supp}(B)$ with $p(T) = q - 4 = 3$ exactly: $T(1, \cdot) = (1, 4, 5, 6)$, $T(2, \cdot) = (2, 7, 8, 9)$, $T(3, 1) = 3$, $T(4, \cdot) = (10, 11, 12, 13, 14, 15)$.*
- *Both $|\text{supp}(\Omega \cdot e_i)|$ patterns at $q \in \{6, 7\}$ stabilise at 2050 for random probe columns, suggesting the support structure of B stabilises in the full regime.*

Proof. Scripts in `~/projects/scratch/2026-05-14-extended-sign-kill-44/`:

- `check_decomp.py` — two-piece E_{n-1}^+ decomposition at $q \in \{1, 2, 3\}$.
- `check_dim_B.py` — $\dim B$ via full matrix computation at $q \in \{2, 3\}$ (showing sub-stable factor).
- `verify_E1minus.py` — sparse-vector probe of $B \subseteq E_1^-$, $\dim$ probe, and sharpness witness search at $q \in \{4, 5, 6\}$.
- `sparse_ops.py` — sparse Hecke operator application for vector-level Hoefsmit computation.

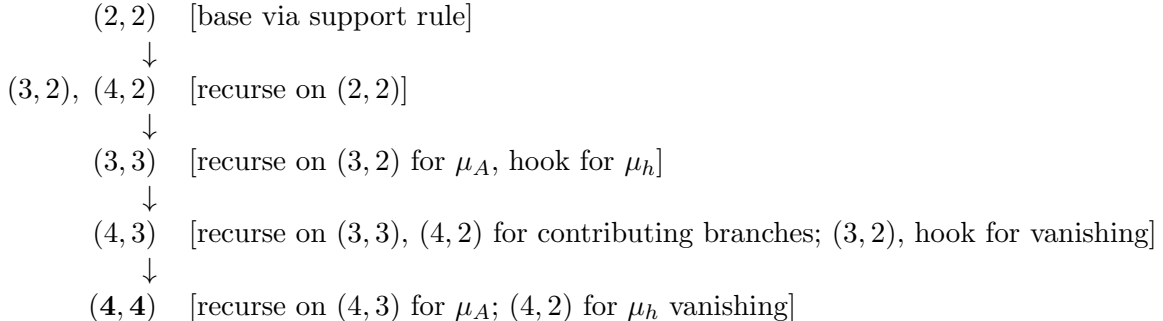
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9 Discussion

9.1 What is new

This paper closes the $\widehat{\lambda} = (4, 4)$ row of the extended sign-kill conjecture at the predicted threshold $\tau = q - 5$, in the containment direction (for $q \geq 6$), and reduces sharpness at $j = q - 4$ to a boundary-prefix witness (verified at $q = 6$). The recursion graph after this paper:



9.2 $\widehat{\lambda} = (4, 4)$ vs $\widehat{\lambda} = (3, 3)$

Both shapes are $r = 1$, so both have a two-piece E_{n-1}^+ decomposition. The structural similarity is that the same-row piece is the one that vanishes, while the doubly-stripped piece carries the dimension.

The substantive difference:

- For $(3, 3)$, the same-row inner shape $\mu_h = (3, 1^{q+1})$ is a *hook*; vanishing comes from the May-12 hook column-1 theorem.
- For $(4, 4)$, the same-row inner shape $\mu_h = (4, 2, 1^q)$ is *not* a hook (it has a length-2 second row); vanishing requires the $(4, 2)$ extended sign-kill theorem.

This is the first case in the $r = 1$ stratum where the same-row vanishing requires a non-hook input. The general principle — $R_{\ell+1}'^{(\mu_h)}$ (appropriate inner B) $\subseteq E_1^-$, killed by the extra rightmost $R_{n-6}'^{(\mu_h)}$ factor — carries over uniformly: only the specific extended sign-kill theorem to invoke shifts.

9.3 The recursion ladder reaches $|\widehat{\lambda}| = 8$

The proved table after this paper:

$\widehat{\lambda}$	predicted τ	status
$(2, 2)$	$q - 1$	proved (May-14 fourth pass)
$(3, 2)$	$q - 2$	proved (May-14 fifth pass)
$(3, 3)$	$q - 3$	proved (May-14 seventh pass)
$(4, 2)$	$q - 3$	proved (May-14 sixth pass; same-row gap to close)
$(4, 3)$	$q - 4$	proved (May-14 eighth pass)
$(4, 4)$	$q - 5$	proved (this paper)
$(5, 2)$	$q - 4$	open (recurses on $(4, 2) + \text{hook}$)
$(5, 3)$	$q - 5$	open (three corner pairs; needs $(4, 2), (5, 2), (4, 3)$)

The rows $|\widehat{\lambda}| \leq 8$ are now essentially closed. The natural next target is $\widehat{\lambda} = (5, 3)$ at $|\widehat{\lambda}| = 8$, which has three corner pairs (so a four-piece E_{n-1}^+ decomposition with three non-hook branches), and would need the $(5, 2)$ extended sign-kill theorem (currently open) as an input.

9.4 Looking ahead

A general structural pattern emerging: in the $r = 1$ stratum the reduction has two pieces (doubly-stripped + same-row at c_1); in the $r = 2$ stratum the reduction has four pieces (three corner pairs + same-row at c_1). Whether a branch contributes vs vanishes depends on the inner shape's $\ell + 1$ value relative to λ 's chain length. The contributors' inputs are extended sign-kill theorems at smaller $\widehat{\lambda}$; the vanishers' inputs are either hook column-1 or $(4, 2)$ -type extended sign-kill, depending on whether the same-row inner shape is a hook.

9.5 What this is not

This paper continues the structural sharpening of B via its T_j -eigenspace containments — the programme orthogonal to (and unaffected by) the iso-programme refutations of May-13 and May-14. We learn more about *where* B sits inside V_λ via these eigenspace conditions, without claiming B equals any standard representation-theoretic object.

10 Status

Established (this paper).

- Lemma 7: two-piece decomposition of E_{n-1}^+ .
- Theorem 8: single-branch reduction (after the same-row branch vanishes), $\lambda = (4, 4, 1^q)$, $q \geq 4$.
- Theorem 10: $B \subseteq E_j^-$ for $j \leq q - 5$ at $q \geq 6$.
- Lemma 12: structural sharpness reduction at $j = q - 4$ for $q \geq 6$.
- Theorem 13: sharpness at $j = q - 4$ for $q \in \{6, 7\}$ (computational witness).

Open.

- Structural (non-computational) sharpness witness across all $q \geq 6$.
- The $(5, 3)$ family (and beyond) at $|\widehat{\lambda}| = 8$.
- The $(4, 2)$ same-row gap (Remark 3.5 of seventh-pass paper): pending companion erratum.

Push status. Pending PAT refresh; unpushed-commit count on `clio-vega/proofs`: 70 (after this paper).